Scapy: Class Participation.

CS6903: Network Security, 2025-26
Department of Computer Science and Engineering
Indian Institute of Technology Hyderabad
Submission Deadline: April 10th, 2026, 11:59 PM, NO EXTENSION

Objective: This assignment is designed to provide hands-on exercises using the Scapy library in Python. You will learn packet crafting, sniffing, network scanning, attack simulation, and detection techniques. Each task is designed to be completed within 2–3 hours. For better understanding, you are encouraged to complete all the tasks, but you need to submit only 1 or 2, depending on whether you have participated in the class presentation. You can see the assigned task here. Each task carries 2 marks.

General Instructions:
- Use Python and the Scapy library for all tasks
- Run scripts with root/administrator privileges
- Perform attack-related tasks only in controlled environments
- Document observations for each experiment

Task 1: ICMP Ping Tool
- **Objective:** Implement a ping tool using ICMP
- **Description:** Send ICMP Echo Request packets and measure round-trip time
- **Expected Output:** Display reachability and RTT
- **Extension:** Compute average RTT over multiple packets

Task 2: Packet Counter Sniffer
- **Objective:** Analyze protocol distribution
- **Description:** Capture packets and classify them into TCP, UDP, and ICMP
- **Expected Output:** Display counts and percentages
- **Extension:** Visualize results using a chart

Task 3: MAC Resolver
- **Objective:** Resolve MAC from IP
- **Description:** Send ARP request and capture reply
- **Expected Output:** Display IP-MAC mapping
- **Extension:** Scan entire subnet

Task 4: TCP Flag Inspector

- **Objective:** Understand TCP flags
- **Description:** Sniff packets and count SYN, ACK, FIN
- **Expected Output:** Display counts
- **Extension:** Track flag combinations

Task 5: DNS Query Printer

- **Objective:** Extract DNS queries
- **Description:** Capture DNS packets and print queried domains
- **Expected Output:** List domain names
- **Extension:** Count frequency

Task 6: TCP SYN Scanner

- **Objective:** Scan TCP ports
- **Description:** Send SYN packets and analyze responses
- **Expected Output:** Identify open ports
- **Extension:** Add threading for speed

Task 7: UDP Scanner

- **Objective:** Scan UDP ports
- **Description: Send** UDP packets and detect ICMP unreachable
- **Expected Output:** List open/filtered ports
- **Extension:** Handle timeout cases

Task 8: HTTP Sniffer

- **Objective:** Analyze HTTP traffic
- **Description:** Capture and extract URLs
- **Expected Output:** Display visited URLs
- **Extension:** Extract headers

Task 9: Packet Logger

- **Objective:** Store traffic data
- **Description:** Log packet summaries with timestamps
- **Expected Output:** Generate log file
- **Extension:** Export to CSV

Task 10: ARP Spoof Detector

- **Objective:** Detect ARP attacks
- **Description:** Monitor ARP replies for inconsistencies
- **Expected Output:** Raise alerts
- **Extension:** Log suspicious events

Task 11: DNS Spoofing

- **Objective:** Understand DNS attacks
- **Description:** Intercept and spoof DNS response
- **Expected Output:** Redirect domain
- **Extension:** Support multiple domains

Task 12: ICMP Sweep

- **Objective:** Discover live hosts
- **Description:** Ping subnet range
- **Expected Output:** List active hosts
- **Extension:** Parallelize scanning

Task 13: Traceroute

- **Objective:** Trace packet path
- **Description:** Send packets with increasing TTL
- **Expected Output:** Display hops
- **Extension:** Measure latency per hop

Task 14: Port Scan Detector

- **Objective:** Detect scanning behavior
- **Description:** Monitor multiple port attempts
- **Expected Output:** Alert suspicious IP
- **Extension:** Add threshold tuning

Task 15: Protocol Analyzer

- **Objective:** Analyze traffic composition
- **Description:** Calculate protocol percentages
- **Expected Output:** Display distribution
- **Extension:** Plot graph

Task 16: Top Talkers

- **Objective:** Identify active hosts
- **Description:** Count packets per IP
- **Expected Output:** Display top senders
- **Extension:** Include receivers

Task 17: Packet Size Analyzer

- **Objective:** Analyze packet sizes
- **Description:** Compute min, max, and average size
- **Expected Output:** Display statistics
- **Extension:** Plot histogram

Task 18: Connection Tracker

- **Objective:** Track TCP sessions
- **Description:** Monitor SYN, ACK, FIN packets
- **Expected Output:** Identify session lifecycle
- **Extension:** Track duration

Task 19: Suspicious Port Alert

- **Objective:** Detect unusual ports
- **Description:** Monitor traffic for suspicious ports
- **Expected Output:** Generate alerts
- **Extension:** Make a configurable list

Task 20: Flow Count Analyzer

- **Objective:** Analyze network flows
- **Description:** Group packets using 5-tuple (src IP, dst IP, src port, dst port, protocol)
- **Expected Output:** Display the number of flows and packets per flow
- **Extension:** Export flow table for analysis

Submission Format:

- Submit a **tar-ball/zip** file of your assignment on Moodle named as your **roll number** (e.g., cs.tar.gz) containing:
    - Source code for each assigned task. Name each file according to the task, for example: task1.py, task2.py.
    - Include a README.md file. The first line should state whether you participated in the class presentation, followed by an acknowledgement section, including disclosure of sources and any potential discussion with others.
    - An Observations.pdf file describing your observations during development and after execution. You may also include relevant screenshots if needed.

Note: Submission will only be accepted through Moodle. Submission via mail, chat, or any other means will not be entertained.

All the best. Looking forward to the submissions!!... PS: Start Early.